THE ODIN PROJECT · ADVANCED HTML AND CSS

Advanced HTML and CSS



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01

Advanced HTML and CSS: what I covered

Animation

Transforms

Transitions

Keyframes

Accessibility

Introduction to Web Accessibility

WCAG

Semantic HTML

Accessible Colors

Keyboard Navigation

Meaningful Text

WAI-ARIA

Accessibility Auditing

Responsive Design

Introduction to Responsive Design

Natural Responsiveness

Responsive Images

Media Queries

Project: Homepage

02

Transforms and transitions

transform – move, rotate, scale or skew an element without changing the layout

Functions – translate(), rotate(), scale(), skew(), and 3D versions

transition – animates a change between two states, like on :hover

Parts – property, duration, timing function (ease) and delay

Performance – animate transform and opacity, they are cheap for the browser

animation.css

```
.card {
  transition: transform 0.25s ease,
              box-shadow 0.25s ease;
}

/* lift the card on hover */
.card:hover {
  transform: translateY(-6px);
  box-shadow: 0 12px 24px rgb(0 0 0 / 0.15);
}

/* small rotation on the logo */
.logo:hover {
  transform: rotate(-6deg);
}
```

03 Keyframes

@keyframes – define the steps of an animation with from / to or percentages

animation – name, duration, timing, delay, iteration-count, direction

Difference – a transition needs a trigger, keyframes can run on their own and loop

fill-mode – keeps the styles of the first or last keyframe

Use it carefully – too much movement can be distracting

keyframes.css

```css
@keyframes bounce {
  0%   { transform: translateY(0); }
  50%  { transform: translateY(-20px); }
  100% { transform: translateY(0); }
}

.ball {
  animation-name: bounce;
  animation-duration: 1s;
  animation-timing-function: ease-in-out;
  animation-iteration-count: infinite;
}

/* shorthand */
.ball { animation: bounce 1s infinite; }
```

04

Web accessibility and WCAG

Why accessibility

Many people use the web with a screen reader, only the keyboard, zoom or with colour blindness. An accessible site works for all of them, and it is also better for SEO.

WCAG

The Web Content Accessibility Guidelines. Four principles (POUR): Perceivable, Operable, Understandable, Robust, and three levels A, AA and AAA. Most sites aim for AA.

Semantic HTML

Use the right element: <button> for actions, <a> for links, <nav>, <main>, <header>, headings in order. Screen readers understand the page from these elements.

Accessible colours

Text needs enough contrast with the background: at least 4.5:1 for normal text (AA). Never use only colour to show information, like errors.

05

Keyboard navigation and meaningful text

Keyboard – everything must work with Tab, Enter and Space

Focus styles – never remove the outline without a better :focus-visible style

Skip link – lets keyboard users jump over the navigation

Meaningful text – links like "View Library code" instead of "click here"

alt text – describe what the image shows; empty alt for decoration

ally.html

```
<!-- first element in the body -->
<a class="skip-link" href="#main">Skip to
content</a>

.skip-link {
  position: absolute;
  top: -100px;          /* hidden above */
}
.skip-link:focus {
  top: 0;               /* visible on Tab */
}

a:focus-visible {
  outline: 3px solid var(--accent);
}

<img src="library.jpg"
     alt="Library app with book cards">
```

06

WAI-ARIA and accessibility auditing

What ARIA is

Attributes that give extra information to screen readers: role, aria-label, aria-labelledby, aria-hidden, aria-expanded, aria-live.

First rule of ARIA

Don't use ARIA if a native HTML element can do it. A <button> is better than a . No ARIA is better than bad ARIA.

In my Homepage

aria-labelledby connects each section to its heading, aria-label names the navigation, and aria-hidden hides decorative SVG icons.

Auditing

Lighthouse in Chrome DevTools, the axe extension and WAVE find many problems automatically. Then I test by hand with the keyboard and zoom.

07

Natural responsiveness and responsive images

Responsive design – one website that works on phone, tablet and desktop

Natural responsiveness – avoid fixed widths; use max-width, %, Flexbox and Grid

Viewport meta tag – required, or phones show a zoomed-out desktop page

Images – max-width: 100% so they shrink with the container

srcset and sizes – the browser downloads the right image size

responsive.html

```
<meta name="viewport" content="width=device-
width">

img { max-width: 100%; height: auto; }

/* cards wrap by themselves, no media query */
.projects {
  display: grid;
  grid-template-columns:
    repeat(auto-fit, minmax(280px, 1fr));
}

<img src="library-800.jpg"
     srcset="library-400.jpg 400w,
             library-800.jpg 800w"
     sizes="(min-width: 960px) 340px, 90vw"
     alt="Library app">
```

08 Media queries

@media – apply CSS only when a condition is true, like a minimum width

Mobile first – write phone styles first, then add min-width queries

Breakpoints – choose them where the design breaks, not for specific devices

Other features – prefers-color-scheme, prefers-reduced-motion, print

My breakpoints – 640px (tablet) and 960px (desktop)

media.css

```
/* phone first: no media query */
.hero-inner { display: grid; }

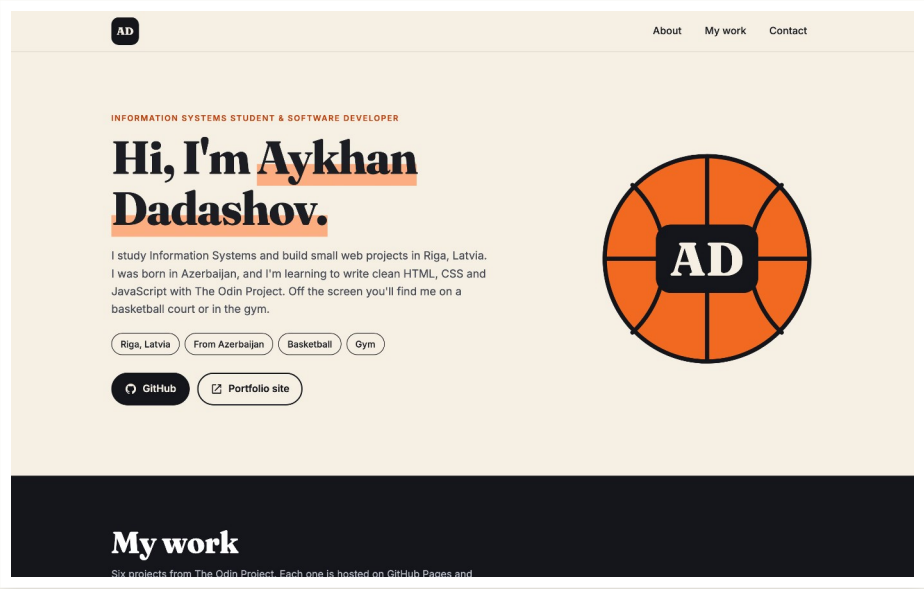
/* tablet and up */
@media (min-width: 640px) {
  .hero { padding: 4rem 0; }
}

/* desktop: text left, portrait right */
@media (min-width: 960px) {
  .hero-inner {
    grid-template-columns: 1fr 340px;
  }
}

@media (prefers-reduced-motion: reduce) {
  * { transition: none !important; }
}
```

09

Project: Homepage



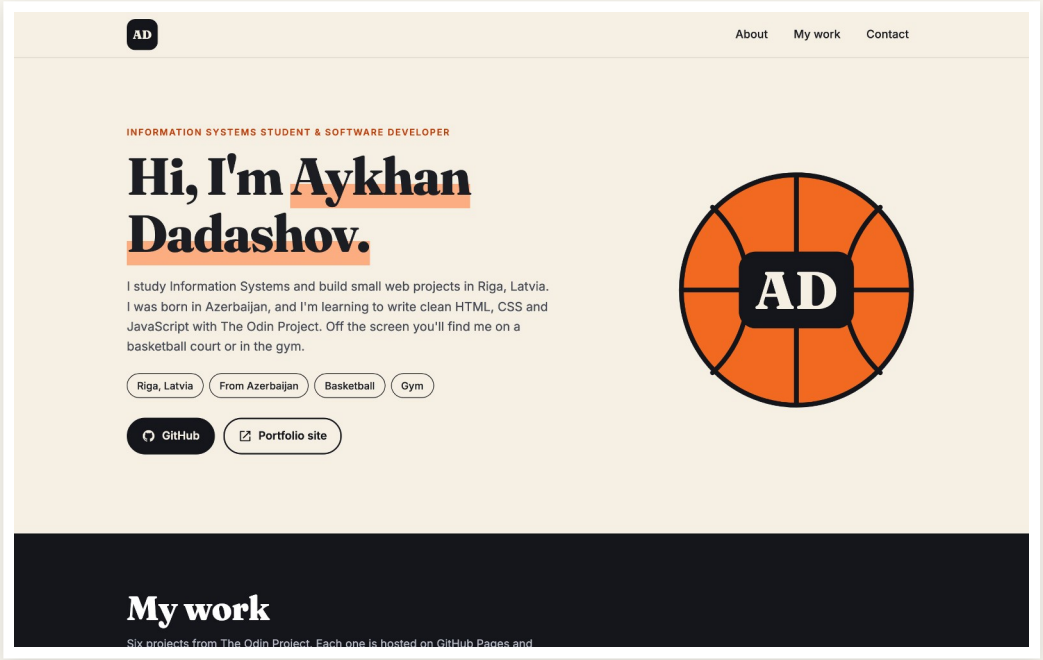
- Personal homepage with my real projects
- Mobile first, breakpoints at 640px and 960px
- Responsive images with srcset and sizes
- Hover transitions on cards and the logo
- Skip link, focus styles, reduced motion

Live ayxan1110.github.io/odin-homepage/

Code github.com/Ayxan1110/odin-homepage

Homepage on phone and desktop





Same HTML, different layout: one column on the phone, two columns on desktop

11 Projects on GitHub

Homepage	ayxan1110.github.io/odin-homepage/	github.com/Ayxan1110/odin-homepage
Profile page	ayxan1110.github.io/	github.com/Ayxan1110/Ayxan1110.github.io

Thank you!